Station-Bat

SKU:K124-B





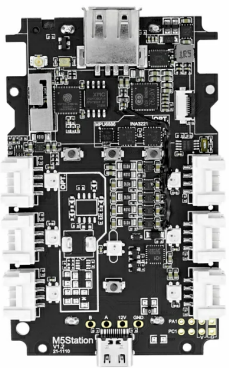
















Description

Station-Bat is a versatile **industrial-grade** programmable embedded controller. It uses the Espressif **ESP32** as the main control chip, integrates a **Wi-Fi** solution, and is equipped with a dual-core low-power **Xtensa® 32-bit LX6** microprocessor with a main frequency of up to **240MHz**. It comes with **16MB Flash**, a **1.14-inch full-color HD IPS display panel** with a resolution of **240×135**, a **physical button panel**, **rich peripherals**, **two sets of six expansion interfaces**, and features **low-power sleep/timed wake-up** functionality. It also integrates the IMU **MPU6886**.

It supports both **USB Type-C** and **internal rechargeable 18650 battery** power supply methods. The onboard high-power-density fully integrated boost DC/DC converter SCT12A0DHKR ensures the **stability** of electrical equipment operation in complex application scenarios.

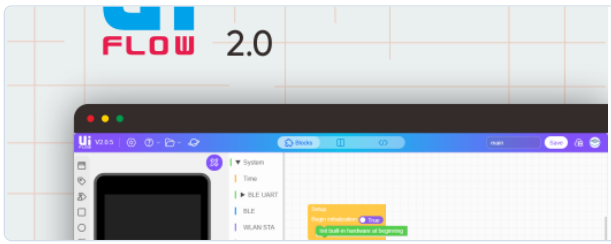
This controller can be used in industrial field control, smart buildings, multi-channel data acquisition nodes, and developer prototype design, among other application scenarios.

Tutorial



UIFlow

This tutorial will introduce how to control the Station-Bat device via the UIFlow graphical programming platform



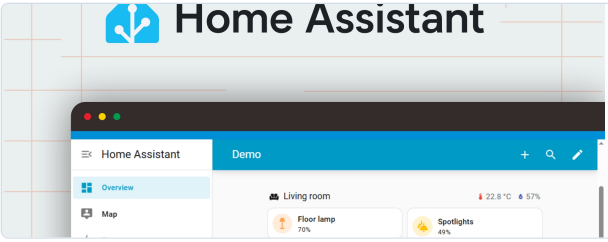
UiFlow2

This tutorial will introduce how to control the Station-Bat device via the UiFlow2 graphical programming platform



Arduino IDE

This tutorial will introduce how to program and control the Station-Bat device using the Arduino IDE.



Home Assistant

This tutorial explains how to integrate Station-Bat into Home Assistant.

Features

- **Interactive Design**:
  - 1.14-inch IPS display panel
  - 6-axis gyroscope MPU6886
  - 3 programmable physical buttons
  - 1 power button
  - 7 programmable RGB LEDs
- **Power Design**:
  - Input:
    - **AXP192** power management chip
  - Output:
    - Each interface (5 Grove, 1 USB-A) uses an electronic switch **SGM2553D** for independent on/off control
    - 6 Grove interfaces use **INA3221** for voltage/current monitoring, and USB-A uses **INA199** for current monitoring
    - Integrated high-power-density fully integrated boost DC/DC converter **SCT12A0DHKR**
  - Low Power:
    - Integrated RTC **BM8563** for low-power sleep/timed wake-up
- **Port Design**:
  - 6 Grove expansion interfaces
    - Port A1/A2 share power and signal pins
    - Port B1/B2/C1/C2 have independent power and signal pins
  - USB Type-A is for power output only, with no signal pins
- **Structural Design**:
  - **DIN rail**
  - **Magnetic**
  - **Wall-mounted**
  - **Screws**
  - **Cable ties**
- Development Platform
  - UiFlow1
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x Station-Bat
- 1 x USB Type-C cable (50cm)
- 1 x Hex key

Applications

- IoT controller
- Multi-channel data acquisition
- IoT product prototype design
- DIY projects

Specifications

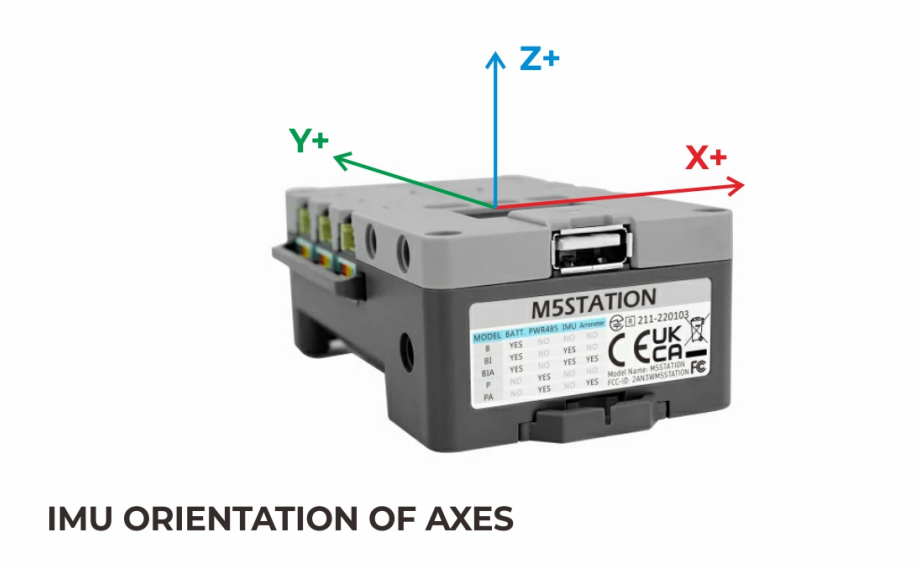
Specification	Parameter
SoC	ESP32-D0WDQ6-V3 @ Xtensa® 32-bit LX6 dual-core processor, clock frequency 240MHz
DMIPS	600
SRAM	520KB
Flash	16MB
Wi-Fi	2.4 GHz Wi-Fi
USB Power	5V@1A
Battery Specification	18650 x 2 (parallel)
Battery Capacity	2200mAh
Host Ports	USB Type-C x 1, HY2.0-4P (I2C+I/O+UART) x 6, Full-Size USB Type-A (OUTPUT)
LED	SK6812 x 7
Buttons	Power button, physical buttons x 3
IPS LCD Screen	1.14"@240 x 135 ST7789V2
IMU	MPU6686
RTC	BM8563
PMU	AXP192
Voltage / Current Sensor	INA3221 + INA199
USB Chip	CH9102F
DC/DC Boost	SCT12A0DHKR
Power Distribution Switch	SGM2553D
Antenna	2.4G 3D antenna
Operating Current	146.3mA
Sleep Current	31.6uA
Operating Temperature	0 ~ 60°C
Base Screw Specification	Hex Cup Head M2 x 8 Screw
Case Material	ABS+PC
Product Size	88.0 x 65.0 x 35.0mm
Product Weight	170.0g
Package Size	104.0 x 73.0 x 54.5mm
Gross Weight	209.4g

Learn

Power On/Off

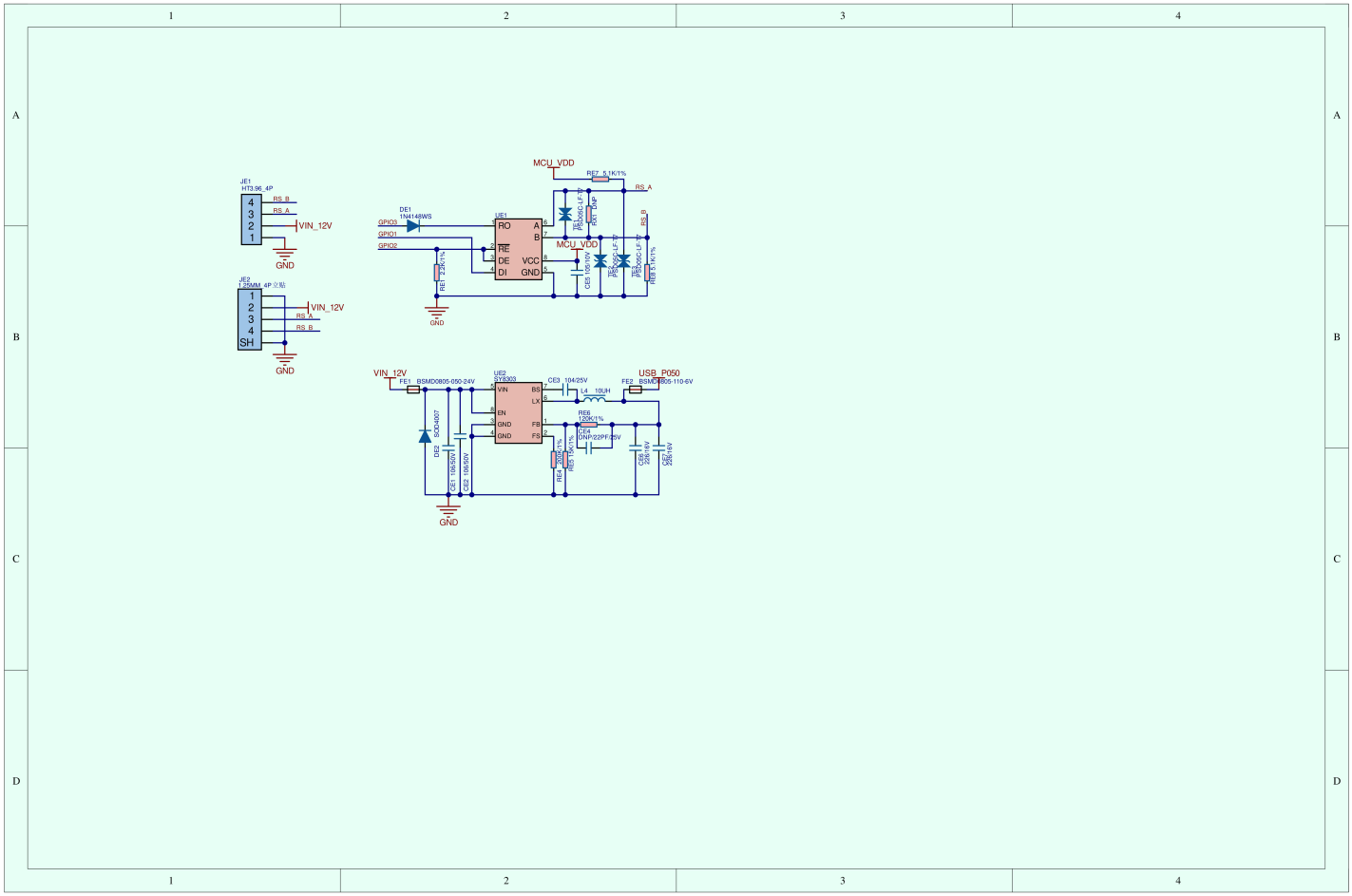
- Power on: Single-click the center power button
- Power off: Press and hold the center power button for 4 seconds

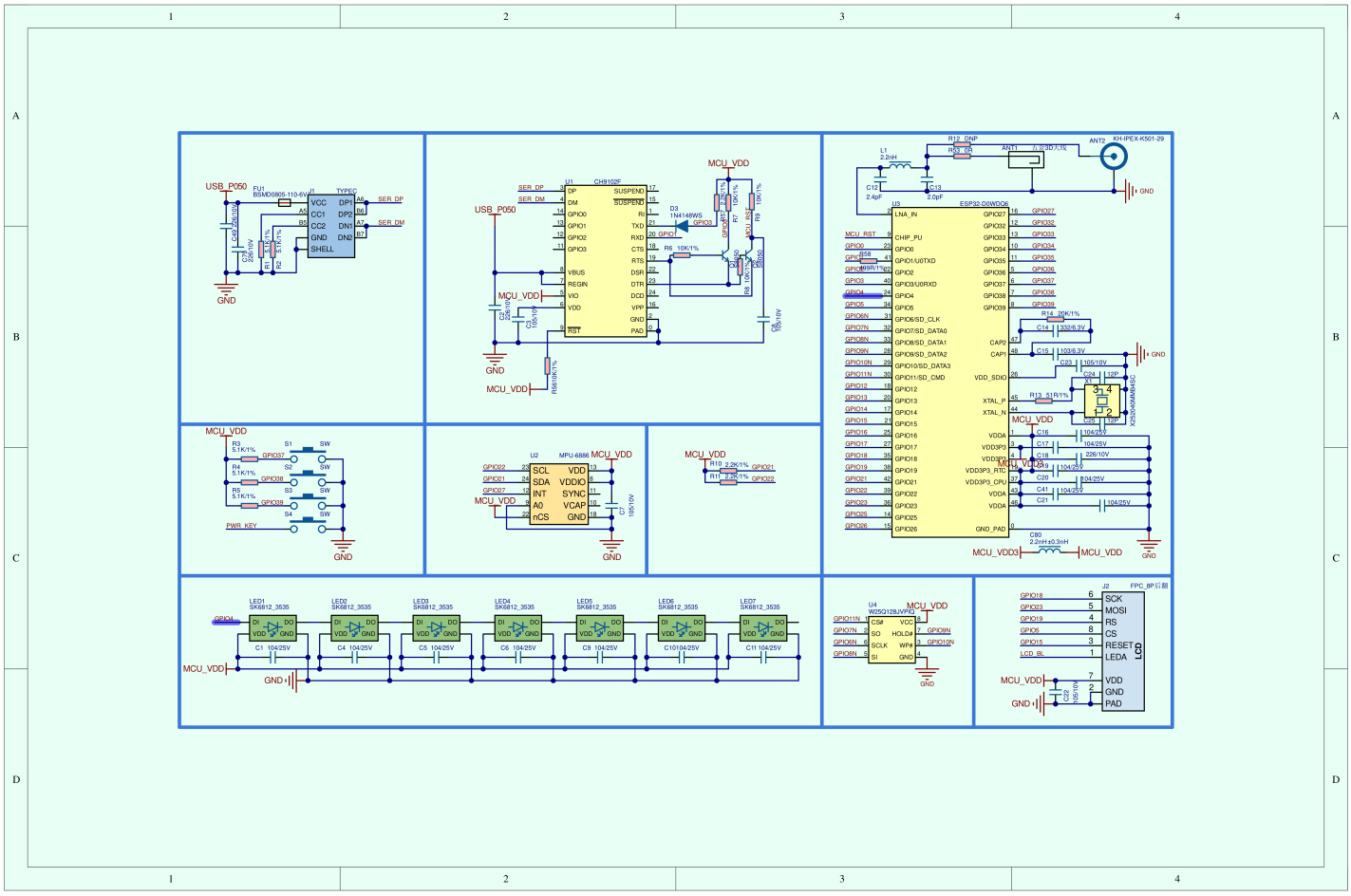
IMU Triaxial Direction Schematic Diagram

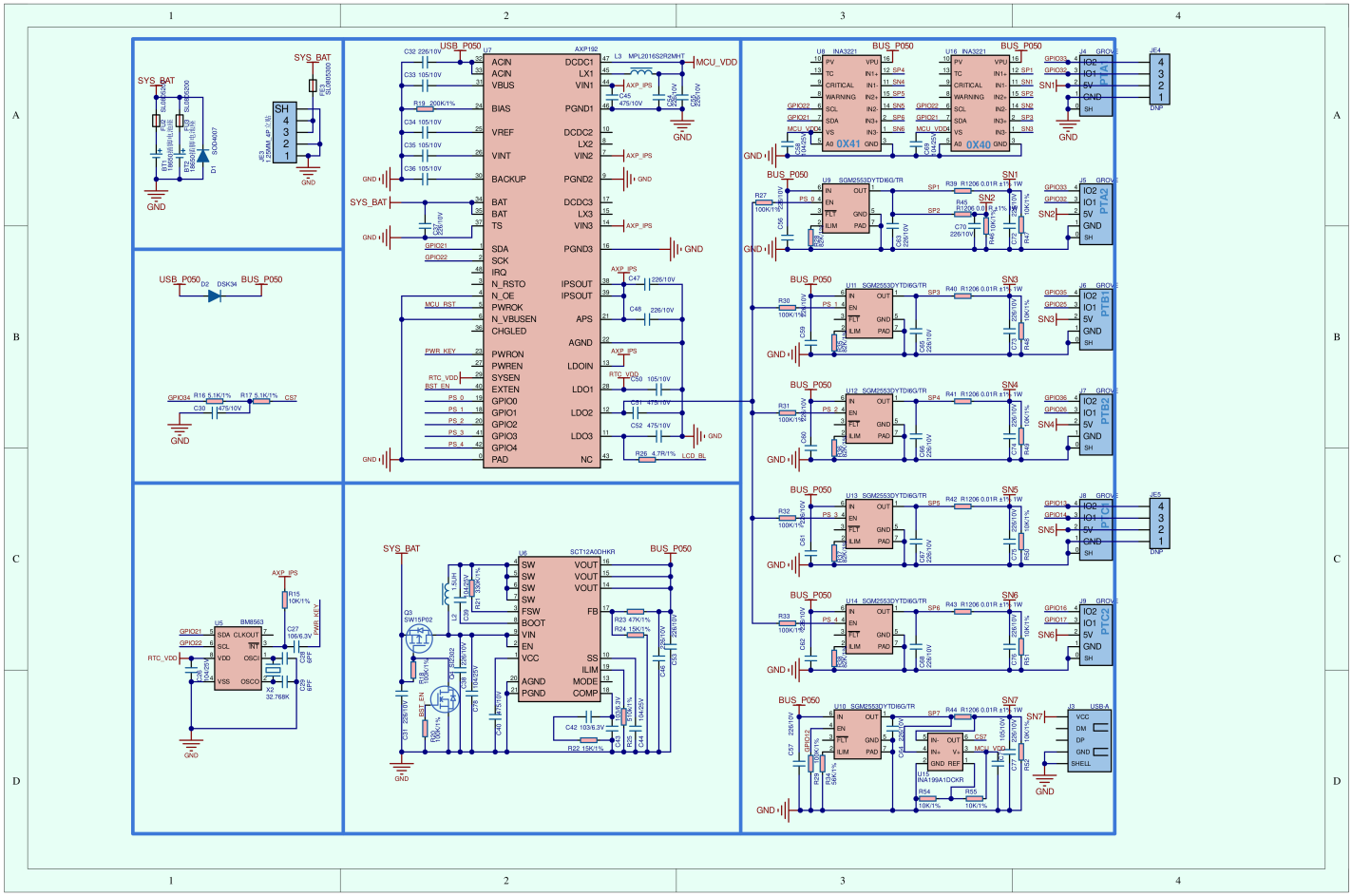


Schematics

- Station-Bat Schematics PDF

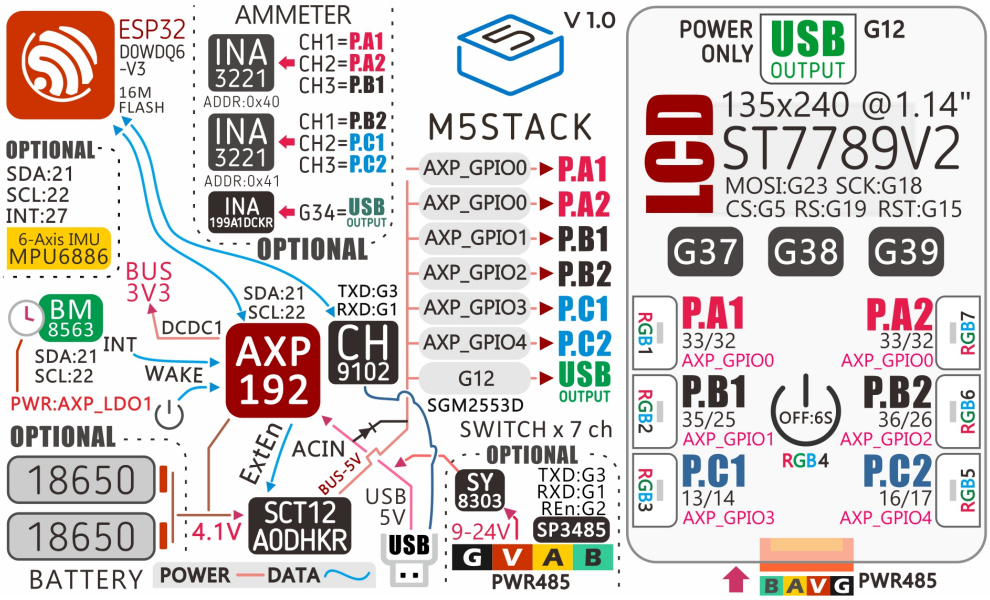

MCU_VDD
VIN_12V
GND
USB_P050
RS_A
RS_B
GPIO1
GPIO2
GPIO3


USB_P050
MCU_VDD
GND
SER_DP
SER_DM
ESP32-D0WDQ6
KH-IPEX-K501-29
PWR_KEY
SCK
MOSI
RS
CS
RESET
LEDA
VDD
GND
PAD



> Schematic updated to v1.3: The 5V power control of the Grove port has been changed from default pull-up enable to control by the LDO2 pin of the AXP192 power management chip.

PinMap



Buttons

ESP32-D0WDQ6-V3	G37	G38	G39
Buttons	BUTTON A	BUTTON B	BUTTON C

Screen

Driver Chip: ST7789v2 Resolution: 135 x 240 @1.14"

ESP32-D0WDQ6-V3	G5	G15	G18	G19	G23
AXP192 Chip					
LCD	CS(Chip Select)	RESET	SCK(SCLK)	RS(Date/Command)	MOSI

AXP192 Chip	AXP_LDO3
LCD	LCD_BL

LED Strip

ESP32-D0WDQ6	G4
hardware	SIG Pin

IMU (3-axis gyroscope + 3-axis accelerometer)

ESP32-D0WDQ6-V3	G21	G22	G27
MPU6886	SDA	SCL	INT

RTC

ESP32-D0WDQ6-V3	G21	G22
BM8563	SDA	SCL
AXP192		

AXP192	AXP_PWR
BM8563	INT

Current/Voltage Monitor

ESP32-D0WDQ6-V3	G21	G22	Control Channel
INA3221(0x40)	SDA	SCL	AXP_GPIO0, AXP_GPIO1
INA3221(0x41)	SDA	SCL	AXP_GPIO2, AXP_GPIO3, AXP_GPIO4

Internal I2C Connection

ESP32-D0WDQ6-V3	G21	G22
MPU6886(0x68)	SDA	SCL
AXP192(0x34)	SDA	SCL
BM8563(0x51)	SDA	SCL
INA3221(0x40、0x41)	SDA	SCL

PWR485

Chip	G3	G1	G2	(DC-DC 9~24->5V)	GND
SP3485	TXD	RXD	REn (READ Enable)		GND
SY8303				VIN_12V	

Power Management Chip (AXP192)

RTC	LCD BackLight	ESP32-3.3V SK6812, INA3221, CH902F
LDO1	LDO3	DC-DC1

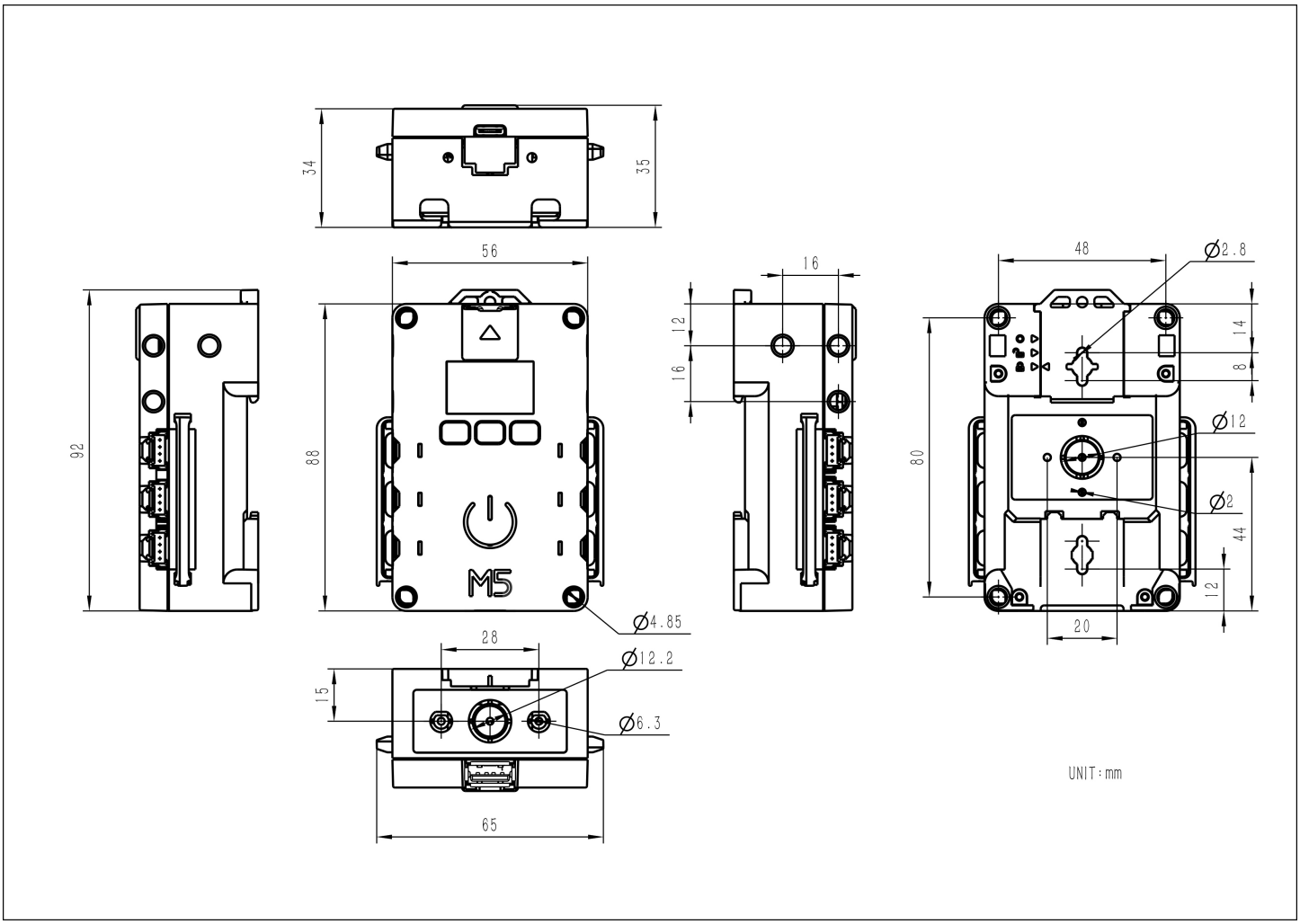
USB to Serial

ESP32-D0WDQ6-V3	G3	G1
CH9102F	TXD	RXD

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A1	GND	5V	G32	G33
PORT.A2	GND	5V	G32	G33
PORT.B1	GND	5V	G25	G35
PORT.B2	GND	5V	G26	G36
PORT.C1	GND	5V	G14	G13
PORT.C2	GND	5V	G17	G16

Model Size



Structure

- Station-Bat Structure Files

Datasheets

- ESP32
- ST7789v2
- SY8303AIC
- SP3485
- SCT12A0DHKR
- INA3221_CN
- INA199_CN
- SGM2553
- BM8563
- MPU6886
- AXP192 Datasheet
- AXP192 Register

Softwares

Quick Start

- Station-Bat Controller Home Assistant Integration

Arduino

- Station-Bat Arduino Quick Start
- Station-Bat Arduino Library

UiFlow1

- Station-Bat UiFlow1 Quick Start

UiFlow2

- Station-Bat UiFlow2 Quick Start

USB Driver

Click the link below to download the driver that matches your operating system. After extracting the compressed file, select the installation package corresponding to your operating system's bit version for installation. (**CH9102_VCP_SER_MacOS v1.7** may report an error during installation, but it is actually installed successfully. Ignore the error.) If you encounter issues with downloading programs (e.g., timeout or "Failed to write to target RAM"), try reinstalling the device driver.

Driver Name	Applicable Driver Chip	Download Link
CH9102_VCP_SER_Windows	CH9102	Download
CH9102_VCP_SER_MacOS v1.7	CH9102	Download

Easyloader

Easyloader	Download Link	Notes
Station-Bat Test Easyloader	Download	/

Video

Product Comparison

Station-485	Station-Bat
Equipped with PWR485 (RS485 + power input)	Onboard MPU6886, supports two 18650 batteries (parallel)

To compare information on the Station series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.